ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 4

# PROJECT MANAGEMENT

## PAPER 1 · PRACTICE A

**MARKS: 100** **TIME: 3 hours**

**NC(V) Information Technology and Computer Science, Level 4. Section A: objective questions (30) · Section B: project selection and the business case, planning and scheduling, cost control and earned value, risk and quality management, procurement, leadership and closure (70).**

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 4. It is not a Department of Higher Education and Training examination paper.

## INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Show ALL formulae and calculations. Round off to TWO decimal places. A non-programmable calculator may be used.
5. Write neatly and legibly.

## SECTION A

## QUESTION 1

**1.1** Various options are given as possible answers to the following questions. Choose the answer and write only the letter (A–D) next to the question number (1.1.1–1.1.10).

1.1.1 The document that justifies why a project should be undertaken, including costs and benefits, is the …

A risk register.
B business case.
C Gantt chart.
D lessons-learnt report.

1.1.2 The time it takes for a project's savings to repay its initial cost is the …

A critical path.
B lead time.
C float.
D payback period.

1.1.3 In earned value management, EV stands for …

A earned value.
B estimated value.
C expected variance.
D extra value.

1.1.4 A cost performance index (CPI) less than 1 means the project is …

A under budget.
B ahead of schedule.
C over budget.
D exactly on budget.

1.1.5 A schedule performance index (SPI) of 1,2 means the project is …

A behind schedule.
B over budget.
C ahead of schedule.
D on schedule.

1.1.6 Transferring a risk to another party, e.g. by buying insurance, is risk …

A transfer.
B avoidance.
C acceptance.
D mitigation.

1.1.7 Checking the quality of deliverables by inspection and testing is quality …

A planning.
B policy.
C assurance only.
D control.

1.1.8 A formal invitation to suppliers to submit prices for supplying goods or services is a …

A project charter.
B request for quotation / tender.
C change request.
D status report.

1.1.9 The leadership style in which the leader makes all decisions alone is …

A democratic.
B laissez-faire.
C servant.
D autocratic.

1.1.10 The stage of team development in which conflict about roles and ideas often occurs is …

A forming.
B storming.
C norming.
D performing.

(10)

**1.2** Indicate whether the following statements are TRUE or FALSE. Choose the answer and write only 'true' or 'false' next to the question number (1.2.1–1.2.10).

1.2.1 A positive cost variance (CV) means the work cost less than planned.

1.2.2 Risk impact × likelihood can be used to rank risks.

1.2.3 Quality assurance focuses on the processes used to produce deliverables.

1.2.4 A fixed-price contract places more cost risk on the client than on the supplier.

1.2.5 A project with a shorter payback period is usually preferred if other factors are equal.

1.2.6 The budget at completion (BAC) is the total approved budget of the project.

1.2.7 A steering committee usually makes the day-to-day technical decisions.

1.2.8 Conflict in a team can sometimes lead to better ideas when it is managed well.

1.2.9 Fast-tracking means doing activities in parallel that were planned in sequence.

1.2.10 A post-implementation review is done before the project starts.

(10)

**1.3** Choose a term from COLUMN B that matches a description in COLUMN A. Write only the letter (A–J) next to the question number (1.3.1–1.3.10).

| | **COLUMN A** | | **COLUMN B** |
|---|---|---|---|
| 1.3.1 | Value of the work planned to be done by a certain date | A | sign-off |
| 1.3.2 | Money actually spent on the work done | B | estimate at completion (EAC) |
| 1.3.3 | Forecast of the total cost of the project at the end | C | return on investment (ROI) |
| 1.3.4 | Grid that rates risks by probability and impact | D | actual cost (AC) |
| 1.3.5 | Changing the plan to remove a risk completely | E | planned value (PV) |
| 1.3.6 | Diagram showing possible causes of a quality problem | F | steering committee |
| 1.3.7 | Group of senior people who oversee and give direction to the project | G | risk matrix |
| 1.3.8 | Buying goods and services from outside suppliers | H | risk avoidance |
| 1.3.9 | Percentage return on the money invested in the project | I | fishbone (Ishikawa) diagram |
| 1.3.10 | Formal acceptance of deliverables by the client | J | procurement |

(10)

**TOTAL SECTION A: [30]**

## SECTION B

**Case study: Online registration system for a TVET college**

A TVET college wants to replace its paper-based student registration with an online system. Every January, thousands of students queue for days. The project budget (BAC) is R400 000 and the planned duration is 10 weeks. An external software company will develop the system, and the college IT department will provide the servers and train staff.

## QUESTION 2: PROJECT SELECTION AND THE BUSINESS CASE

**2.1** Give FOUR benefits of the online registration system that should be included in the business case. (4)

**2.2** The college compares two options. Option A costs R400 000 and is expected to save R160 000 per year. Option B costs R300 000 and saves R90 000 per year.

2.2.1 Calculate the payback period of EACH option. (4)

2.2.2 Calculate the return on investment (ROI) of Option A over five years. (3)

2.2.3 Which option should the college choose? Give TWO reasons, one of which is not financial. (3)

**2.3** Name TWO non-financial criteria used to select projects. (2)

**[16]**

## QUESTION 3: COST CONTROL AND EARNED VALUE

**3.1** At the end of week 5 the following information is available: planned value (PV) R200 000; 40% of the total work has been completed; actual cost (AC) R180 000.

3.1.1 Calculate the earned value (EV). (2)

3.1.2 Calculate the schedule variance (SV) and the cost variance (CV). (4)

3.1.3 Calculate the SPI and the CPI. (4)

3.1.4 Interpret the results: is the project ahead of or behind schedule, and over or under budget? (2)

3.1.5 Calculate the estimate at completion (EAC) using EAC = BAC ÷ CPI. (2)

**3.2** Suggest TWO actions the project manager could take to get the project back on schedule. (2)

**[16]**

## QUESTION 4: RISK, QUALITY AND PROCUREMENT

**4.1** Identify FOUR risks for the registration project. For EACH, rate the probability and impact (high, medium or low) and give a response strategy. (12)

**4.2** Explain the difference between quality assurance and quality control, with an example of each from this project. (4)

**4.3** Give FOUR criteria the college should use to evaluate the software companies' tenders. (4)

**[20]**

## QUESTION 5: LEADERSHIP, COMMUNICATION AND CLOSURE

**5.1** Name and briefly describe the FOUR stages of team development (Tuckman). (8)

**5.2** Draw up a simple communication plan for THREE stakeholders: what information they receive, how and how often. (6)

**5.3** Explain the purpose of a post-implementation review and name TWO questions it should answer. (4)

**[18]**

**TOTAL SECTION B: [70]**

**TOTAL: 100**

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**END OF QUESTION PAPER**